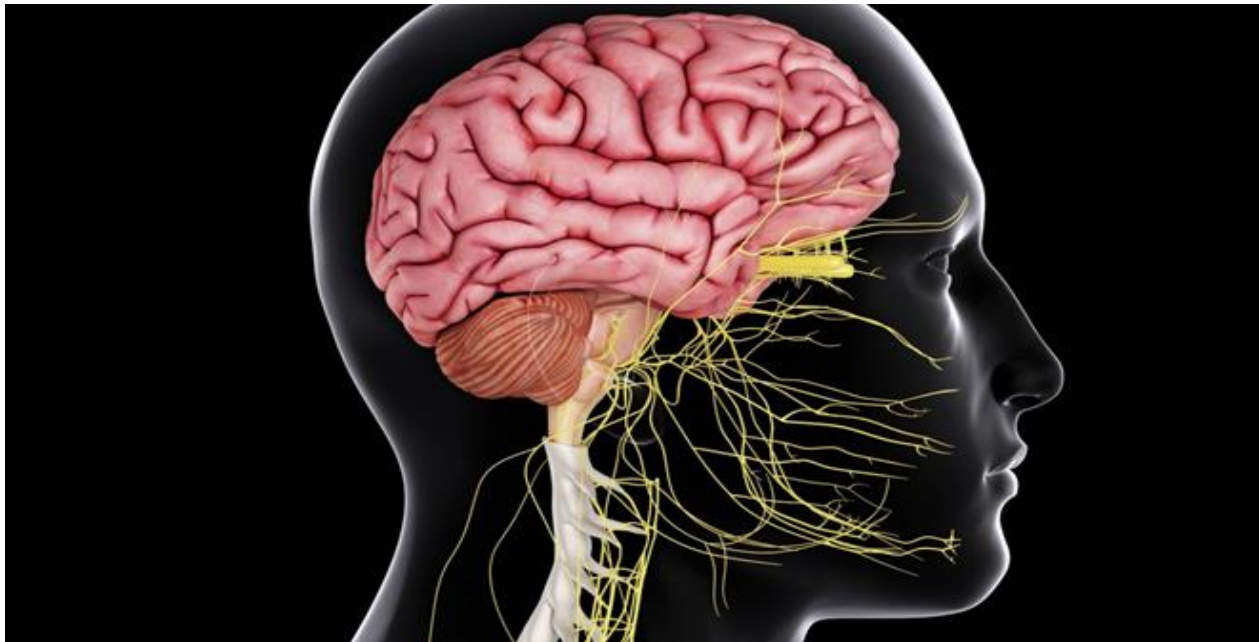


Introduction to Nervous System



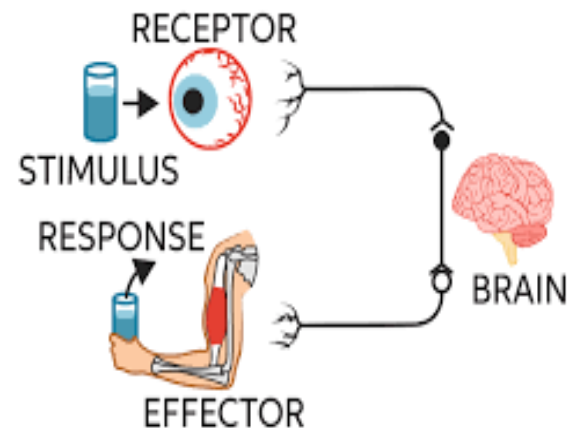
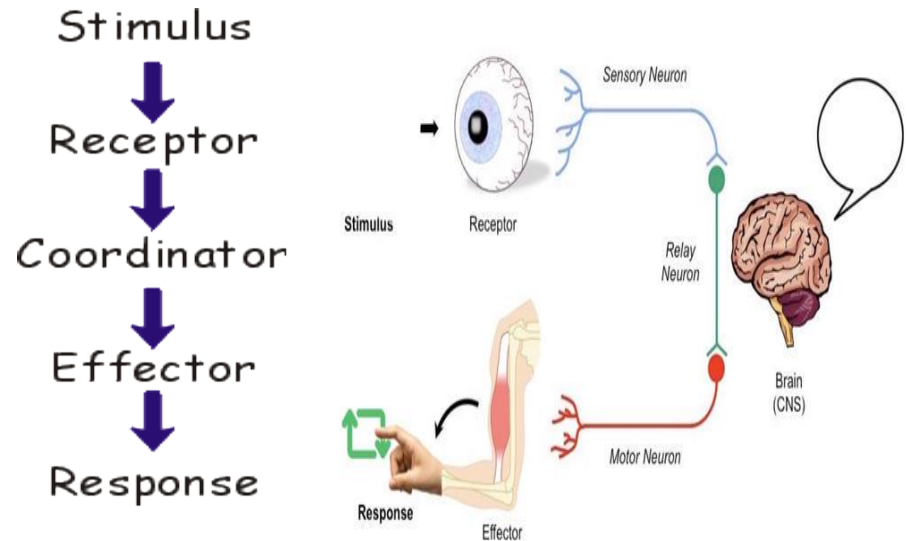
Learning Objectives

At the end of the lecture, the student should be able to:

- **Define nervous system and its divisions.**
- **Describe and classify neurons.**
- **Explain the basic structure of neuron/nerve cell.**
- **Define synapse and its types.**
- **Define gray and white matter.**
- **Describe neuroglia and their types & functions.**

Nervous system:

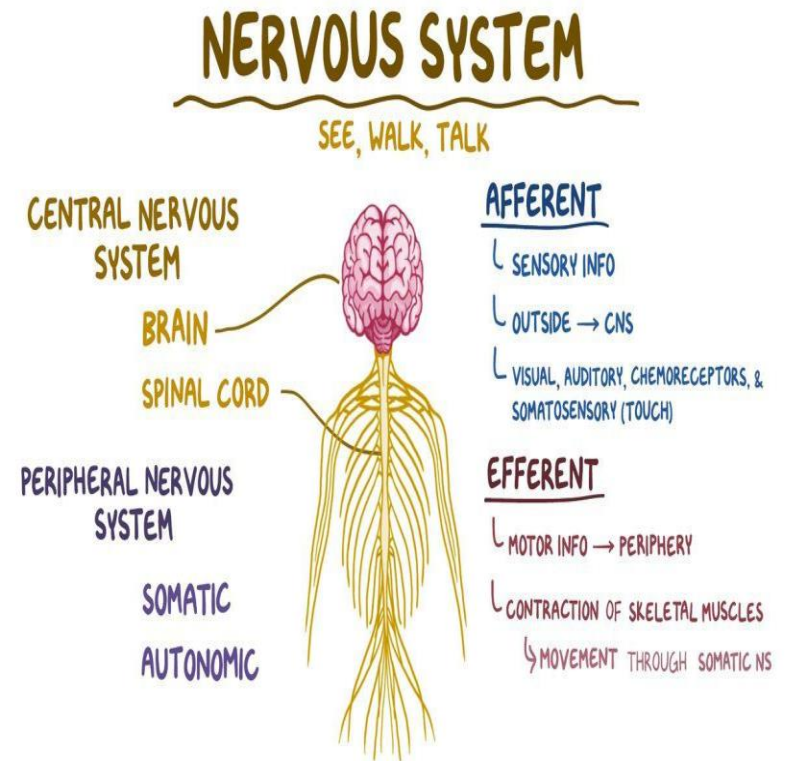
- Nervous system is the chief controlling and coordinating system of the body.
- It is composed of specialized cells, the neurons, who receive sensory stimuli from receptor organs.
- The sensory information is correlated and coordinated within the central nervous system.
- And motor output is then transmitted to the effector organs e.g. muscles, glands etc.



Divisions of nervous system:

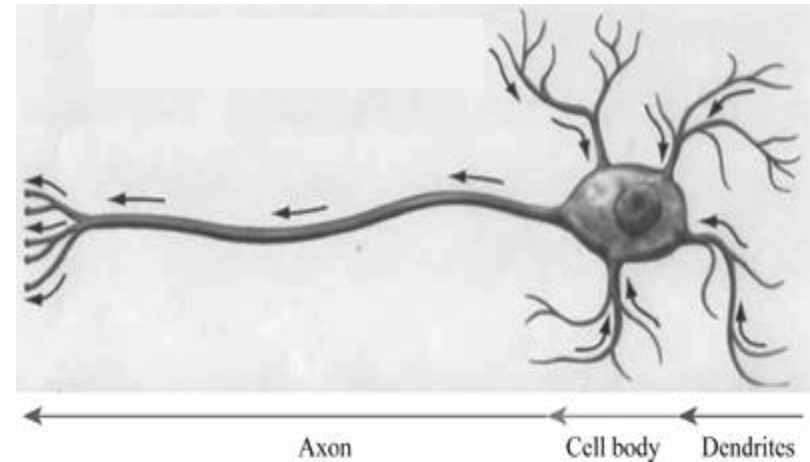
The nervous system is divided into two major parts:

- **Central nervous system:** consists of brain and spinal cord.
- **Peripheral nervous system:** consists of cranial and spinal nerves and their associated ganglia. There are 12 pairs of cranial nerves and 31 pairs of spinal nerves.

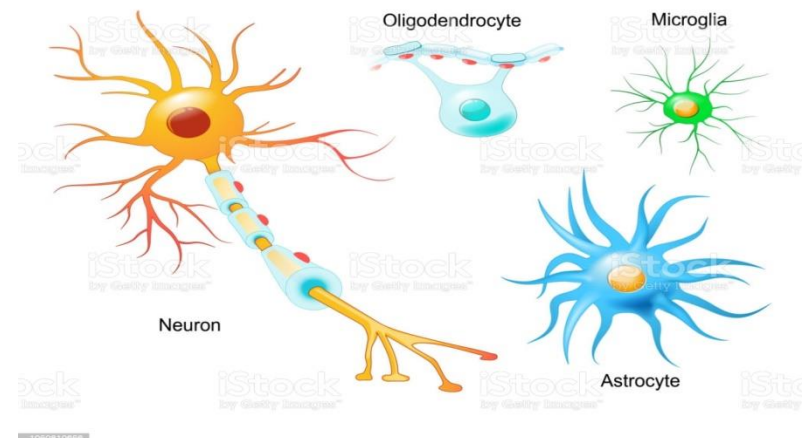


Types of cells of nervous system:

- i. **Neurons:** They are excitable cells specialized for the reception of stimulus and conduction of nerve impulse.
- ii. **Neuroglia:** They are supporting, non-excitable cells which comprise about half the total volume of brain and spinal cord. They are of four types: 1) Astrocytes, 2) Oligodendrocytes, 3) Microglia and 4) Ependyma.

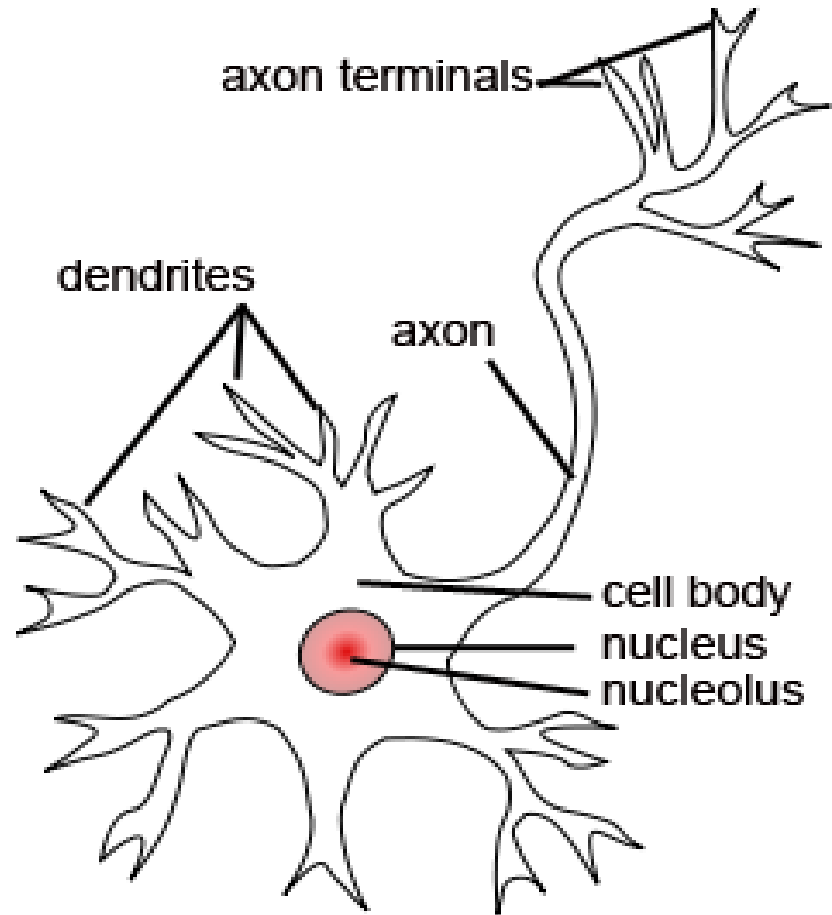


NEURON and GLIAL CELLS



The Neuron:

- Each neuron has a **cell body/perikaryon** and one or more processes called **neurites**.
- The neurites responsible for receiving information towards cell body are called **dendrites**.
- The single long neurite responsible for carrying impulses away from cell body is called **axon**.
- Axons may be myelinated or non-myelinated.



Classification of neurons:

1) According to number of processes/morphology of neurites:

Unipolar /pseudounipolar neurons:

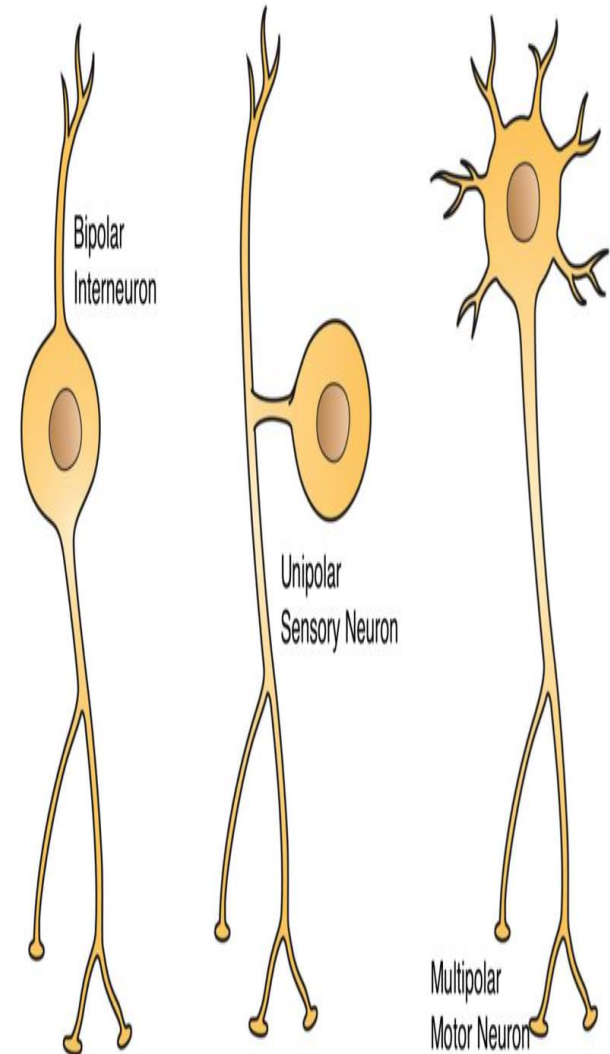
- A single neurite divides into two branches in a T-shaped manner a short distance away from cell body.
- One branch passes towards periphery and functions as dendrites while other passes towards CNS and functions as axon.
- **Examples:** Dorsal root ganglia of spinal nerves and sensory root ganglia of cranial nerves.

Bipolar neurons:

- They have a single axon and a single dendrite arising from opposite poles of fusiform cell body.
- **Examples:** Cochlear and vestibular ganglia, retina and olfactory epithelium.

Multipolar neurons:

- They have a single axon and a number of dendrites arising from cell body.
- **Examples:** Pyramidal cells (cerebral cortex), Purkinje cells (cerebellar cortex) and anterior horn cells (spinal cord).



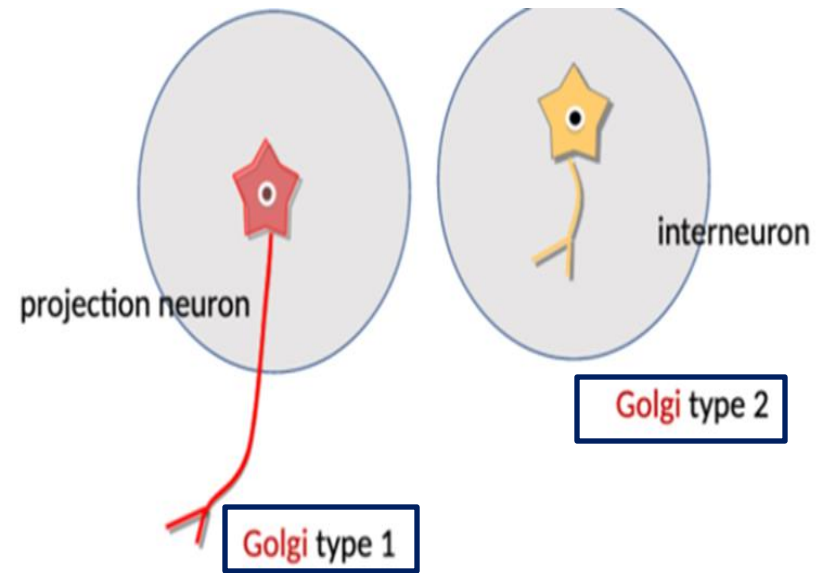
2) According to size:

Golgi type-1 neurons:

- They have a long axon up to 1 m or more.
- Examples: Tracts of brain and spinal cord, nerve fibers of peripheral nerves, Pyramidal cells (cerebral cortex), Purkinje cells (cerebellar cortex) and motor cells of spinal cord.

Golgi type-2 neurons:

- They have a short or no axon.
- They are star-shaped and usually inhibitory in function.
- Examples: Cerebral and cerebellar cortex.





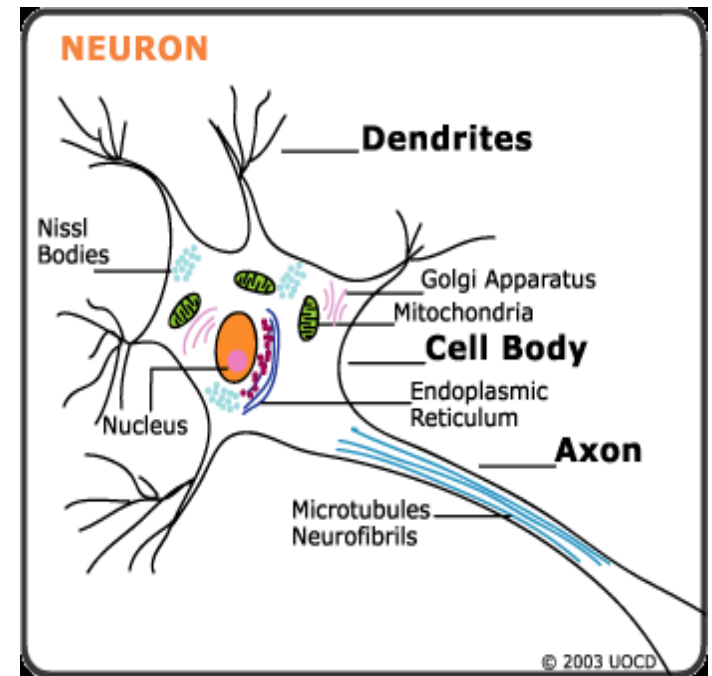
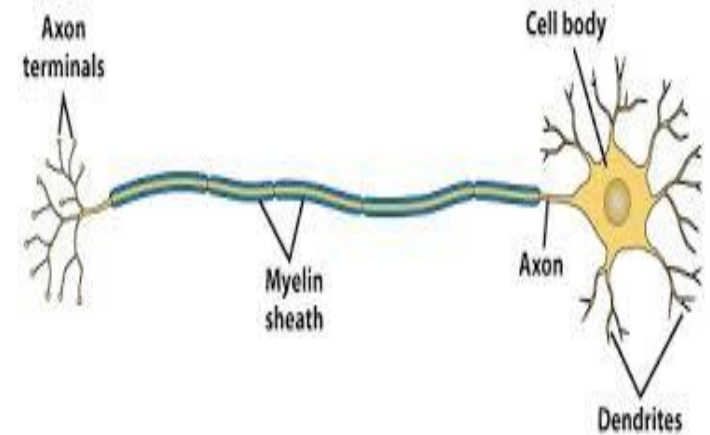
NEURONS CLASSIFICATION :

According to their function :

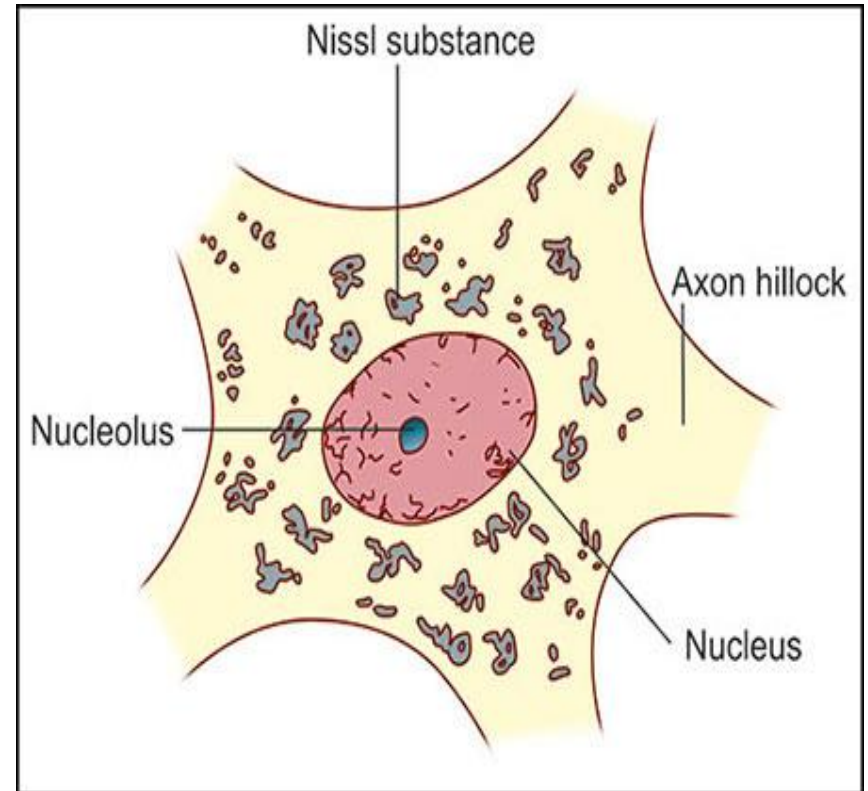
- **Sensory Neuron (afferent)**
 - Receive sensory input → conduct impulses to CNS
- **Motor Neuron (Efferent)**
 - CNS → conduct impulses to muscles, glands and other neurons
- **Interneuron**
 - In the CNS as interconnectors, establish neuronal circuit between sensory and motor neuron

Structure of a neuron:

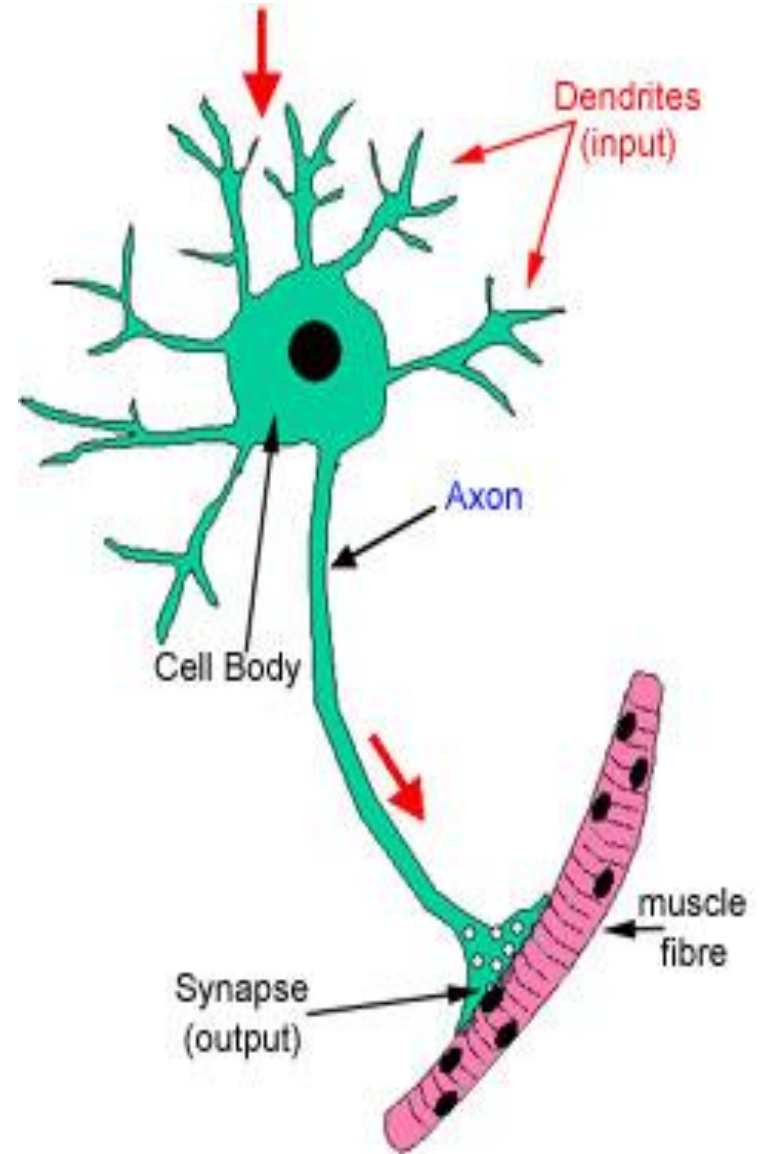
- **The nerve cell body** contains cytoplasm, nucleus and bounded externally by plasma membrane.
- Diameter of cell body may range from 5-135 μm .
- **Nucleus** is central, large, pale-staining and consists of a prominent nucleolus and abundant euchromatin.
- **Cytoplasm** contains organelles such as Nissl substance, golgi apparatus, mitochondria, centrioles, lysosomes, neurofilaments and microtubules and inclusions like lipofuscin and melanin pigments and lipid and glycogen granules.



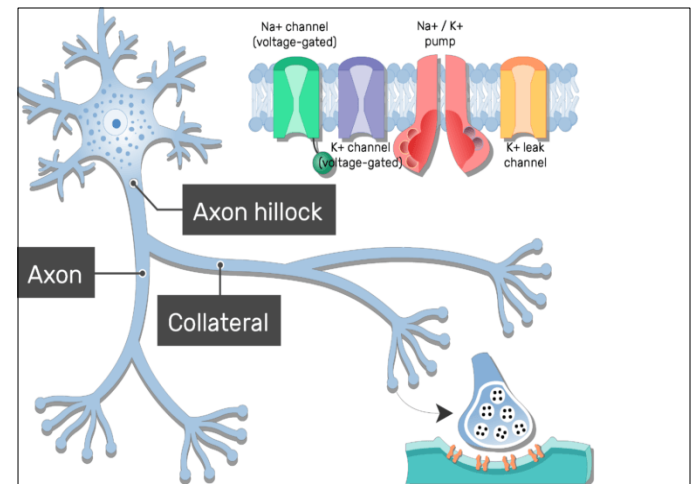
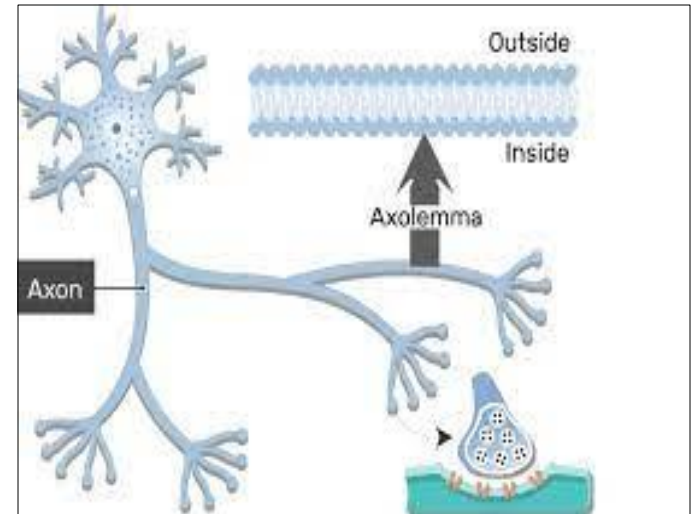
- Nissl substance is made up of rough endoplasmic reticulum arranged in the form of stacked cisternae one on the other.
- **The nerve cell processes/neurites** may be divided into dendrites and axon.
- **The Dendrites** are short processes.



- The dendrites branch profusely.
- Their branches end up in knob-like projections called dendritic spines where synapses are present.
- **The axon** is the longest process of cell body which serves to transmit stimuli to other neurons or effector cells.
- Axon is tubular and uniform in diameter.
- It arises from a small conical elevation on cell body which is devoid of Nissl substance called **axon hillock**

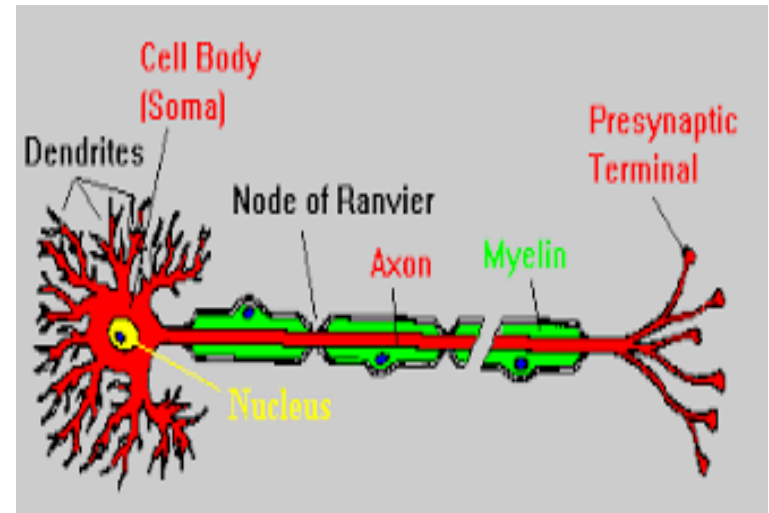
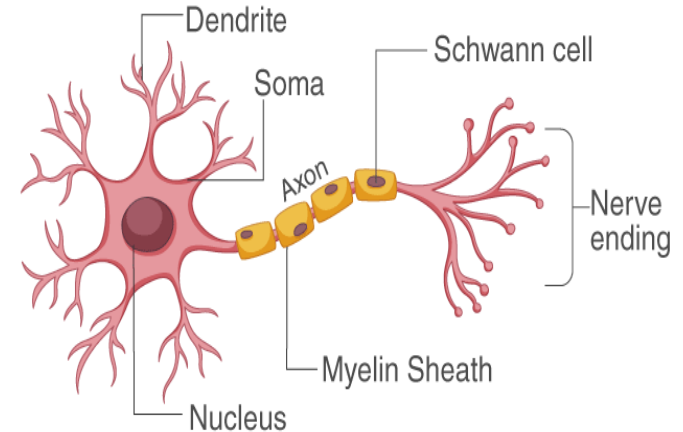


- The initial segment of an axon is the first 50 to 100 micrometers after it leaves the axon hillock of nerve cell body.
- The initial segment is the most excitable of the axon & it is the site at which action potential generates.
- The cytoplasm of axon is termed as **axoplasm** and plasmalemma as **axolemma**.
- It contains mitochondria, smooth ER, microtubules and neurofilaments but no Nissl substance is seen in an axon.
- Axon do not branch profusely but it gives **side-branchings** called **collaterals** and ends by dividing into **terminal branchings** called **telodendria**.



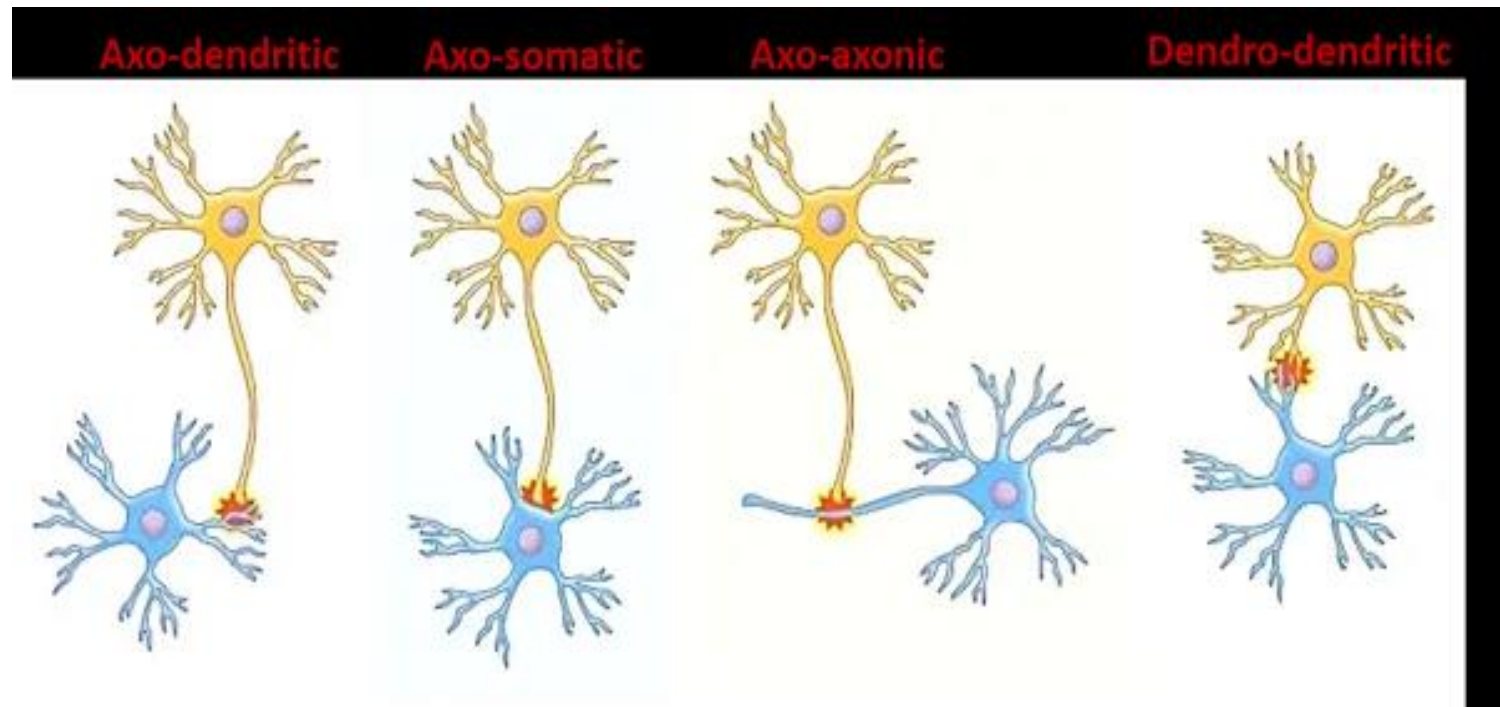
STRUCTURE OF NEURON

- The collaterals and telodendria terminate as small knob-like expansions called **boutons**.
- In a **myelinated** nerve fiber, axon is surrounded by a layer of lipoprotein material called **myelin sheath**.
- Myelin sheath is not a continuous layer and shows interruptions at regular intervals/gaps. These gaps are called **Nodes of Ranvier**.
- Each segment of nerve fiber lying between two consecutive nodes is called **internode**.
- Myelination is carried out by **Schwann cells** in **PNS** and **oligodendrocytes** in **CNS**.



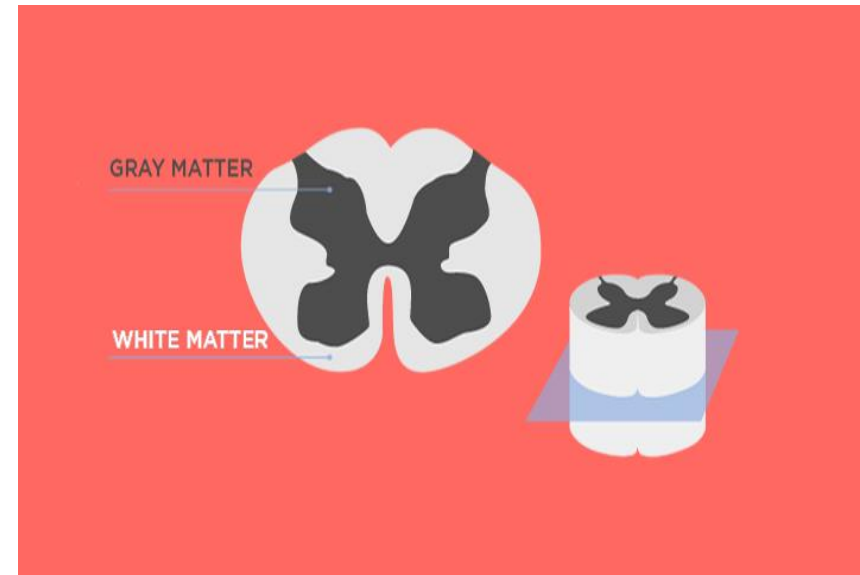
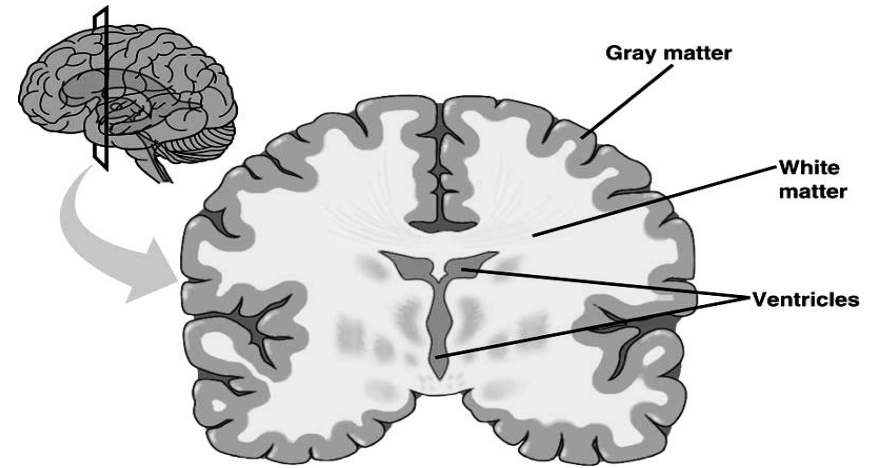
SYNAPSE: When two neurons come in close contact and functional interneuronal communication occurs, the site of such communication is termed as synapse.

- **Types of synapse/synapses:**



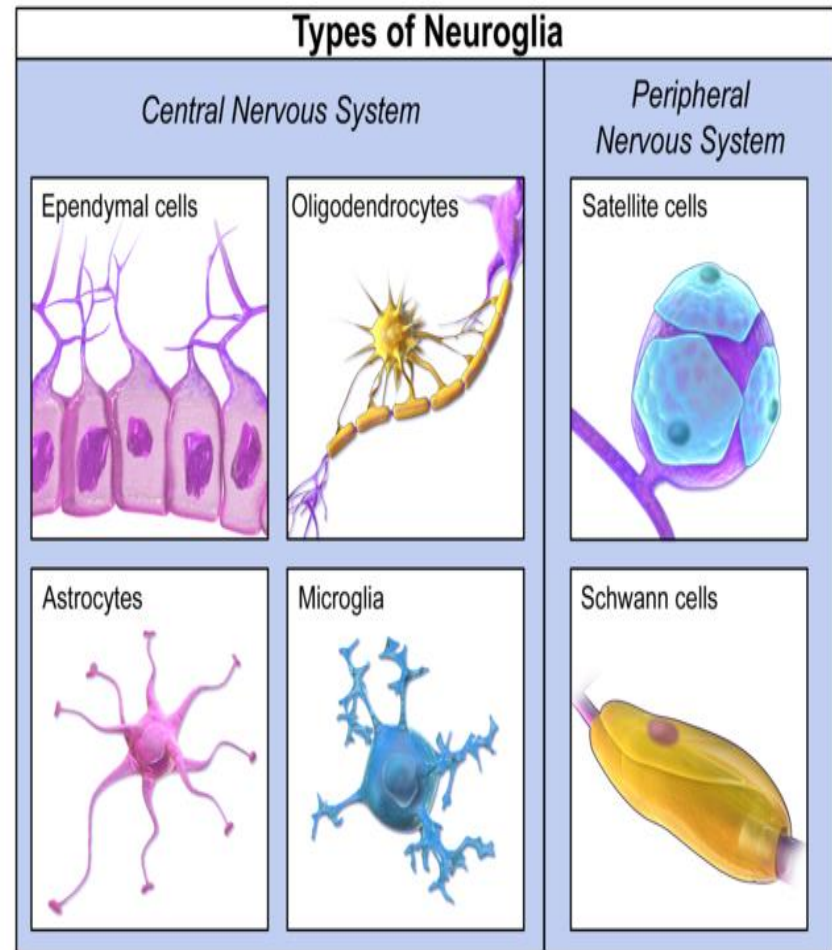
GRAY & WHITE MATTER:

	CNS	PNS
Gray matter (collection of cell bodies)	Nucleus	Ganglion
White matter (collection of myelinated axons)	Tract	Peripheral nerve/nerve



NEUROGLIA:

- Constitute **supporting tissue** of the nervous system.
- They are non-excitabile but provide structural and meta- bolic support to the neurons.
- They are of four main types:
 - I. Astrocytes:** They may be fibrous or protoplasmic and help in establishing and maintaining blood-brain barrirer.
 - II. Oligodendrocytes:** Responsible for myelination of axons in CNS.
 - III. Microglia:** Phagocytosis, in diseased state along with monocytes.
 - IV. Schwann cells:** Myelination of axons in PNS.
 - V. Ependyma:** Line ventricles of brain and central canal of spinal cord and assist in production, circulation and absorption of cerebrospinal fluid.



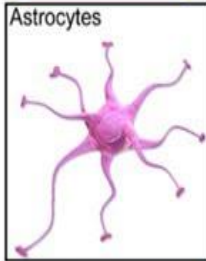
Neuroglial Cell Types & Function



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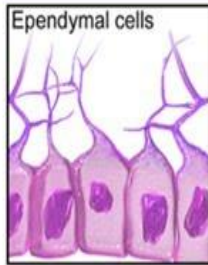
Central Nervous system

Astrocytes



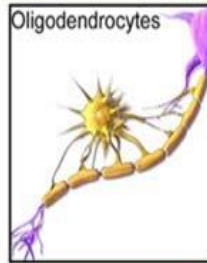
Maintain blood brain barrier
-controlling the levels of neurotransmitter around synapses,
-regulate ion, and providing metabolic support.

Ependymal cells



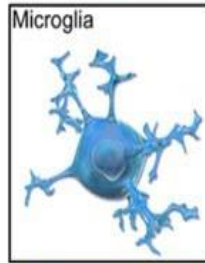
Line spinal cord & ventricles of the brain.
-involved in producing cerebrospinal fluid (CSF).

Oligodendrocytes



Myelinate CNS axons, provide structural framework

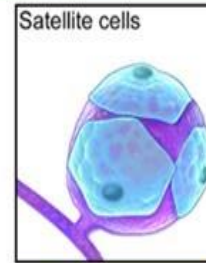
Microglia



Brain's immune cells
-Remove dead cells and pathogens by phagocytosis

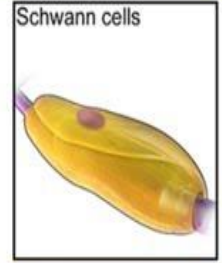
Peripheral Nervous system

Satellite cells



Surround neuron cell bodies in ganglia. Regulate neurotransmitter levels

Schwann cells



Myelinate neurons in PNS. maintenance and regeneration of neurons after injury

References:

- **CLINICAL
NEUROANATOMY**
Richard S. Snell
8th edition
Chapter 2
Pages # 34 - 58

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ANATOMY (DIKIOHS-DUHS)

*Thank
you!*